Main Duct Fan – Maintenance Practices

1. Removal/installation

   A. Remove Main Duct Fan

      1) Remove ejector (refer to 21-40-21, Ejector – Maintenance Practices).

      2) Disconnect fan electrical wiring.

      3) Slacken and remove sleeves and clamps securing ram air fan duct to fan, silencer duct and ventilation duct.

      4) Remove bolts, washers and nuts securing ram air fan duct bracket to flight compartment floor structure.

      5) Move ram air fan duct sufficiently to allow fan to be withdrawn from ram air intake duct. Disconnect ram air valve control at valve handle, if necessary.

      6) Remove bolts and washers securing fan to support brackets.

      7) Remove fan from aircraft.

   B. Install Main Duct Fan

      1) Examine ram air duct to fan seal, change if damaged.

         Note If seal requires changing, bond seal to duct sleeve using Dow Corning A4000 (refer to 20-50-02, Sealant and Adhesives – General Data.

      2) Examine replacement main duct fan for signs of damage.

      3) Position main duct fan in aircraft, ensure inlet end of fan enters ram air duct without damage to seal.

      4) Install bolts and washers to secure fan to support brackets.

      5) Install ram air fan duct and secure to fan, silencer duct and ventilation duct, using sleeves and clamps. Tighten clamps.

      6) Reconnect and adjust ram air valve control, if removed.

      7) Connect fan electrical wiring.

      8) Install ejector (refer to 21-40-21, Ejector – Maintenance Practices).

2. Adjustment/Test

   A. Operational Test

      1) Ensure VENT FAN circuit breaker is in.

      2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

      3) Pull RAM AIR valve control handle to OPEN.

      4) Select VENT FAN switch to ON.

      5) Position hand over windshield heater outlets and check for airflow.

      6) Select VENT FAN switch to OFF.

      7) Push RAM AIR valve control handle to CLOSED.

8) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3. Inspection/Check

A. Check Brushes for Wear

1) Remove main duct fan (refer to Para 1.A).

2) Lift off caps from brush and spring assemblies.

3) Lift out brush and spring assemblies.

Note Mark each assembly and corresponding holder to ensure exact original positioning if brush and spring assemblies do not require replacement.

4) Inspect brush and spring assemblies for brush wear, cracks and chipping.

Note The brushes are to be replaced if cracked, chipped, or when wear exceeds 20% of useful brush length. (Useful brush length (100%) 7/16 inch).

5) Install brush and spring assemblies.

6) Fit caps over installed assemblies.

7) If new brushes are installed, carry out Para 3.A.8) and Para 3.A.9), otherwise omit.

8) Apply 8 to 14 volts to motor terminals and operate fan until brushes are 75 to 90% seated.

WARNING BEFORE ENERGIZING BLOWER OR MOTOR, ASSEMBLY MUST BE SECURED AND GUARD PROVIDED AROUND BLOWER END TO PROTECT PERSONNEL.

9) Increase voltage to 27 volts and operate fan for a further 20 minutes.

10) Install main duct fan (refer to Para 1.B).